

Tobia Poppi

APPLIED SCIENTIST · RESPONSIBLE & MULTIMODAL AI

Modena, Italy

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Professional Summary

PhD Candidate in Artificial Intelligence specializing in **Responsible Generative AI**, **multimodal learning**, and the safety and reliability of Vision-Language Models. Former **Amazon Prime Video Applied Scientist Intern**, where I developed an agentic counterfactual video-generation pipeline accepted at **EMNLP 2026**. Recipient of a **\$30,000 Amazon Post-Internship Research Fellowship** and author of **CVPR Highlight**, **ECCV**, and **ICPR** Oral publications. I build safety-alignment, model-steering, evaluation, and synthetic-data workflows for reliable AI systems.

Research & Industry Experience

Amazon Prime Video

Seattle, USA

APPLIED SCIENTIST INTERN

Jul. 2025 – Nov. 2025

- Designed and implemented an **agentic counterfactual video-generation pipeline** combining VLM reasoning, image-editing models, and video diffusion to target object, action, and temporal hallucinations.
- Generated targeted counterfactual training pairs and evaluated their effect on Video-Language Model hallucinations; the resulting paper was accepted at **EMNLP 2026**.
- Selected as the **inaugural EMEA recipient** of Amazon's Post-Internship Research Fellowship to continue the work through a funded research collaboration.

AlmageLab – University of Modena and Reggio Emilia

Modena, Italy

PHD RESEARCHER

Nov. 2023 – Present

- Develop safety-aware multimodal architectures and empirical evaluations for unsafe generation, retrieval failures, hallucinations, calibration, and robustness, leading to publications at **CVPR**, **ECCV**, **EMNLP**, and **ICPR**.
- Built the **ViSU** multimodal safety dataset and fine-tuned open Llama and CLIP-family models to reduce unsafe outputs while measuring capability retention and downstream utility.
- Lead a joint research track with **Amazon Prime Video** on counterfactual data generation and reliable VLMs, coordinating experiments across academic and industry collaborators.

Hipert Lab – University of Modena and Reggio Emilia

Modena, Italy

RESEARCH FELLOW

Feb. 2022 – Aug. 2022

- Implemented and evaluated geometric deep-learning pipelines for real-time **6-DoF pose estimation**; subsequent work analyzed shortcut bias in fiducial-marker benchmarks and used generative augmentation to improve robustness.

Hipert Lab – University of Modena and Reggio Emilia

Mantova, Italy

RESEARCH FELLOW INTERN

May 2021 – Sep. 2021

- Applied deep-learning tools to data labeling and 3D pose estimation of road objects for the Bachelor's thesis.

Selected Publications

Counterfactual Video Generation for Mitigating Action and Temporal Hallucinations in Video-Language Models – EMNLP 2026

Budapest, Hungary

TOBIA POPPI, BURAK UZKENT, AMANMEET GARG, LUCAS PORTO, GARIN KESSLER, YEZHOU YANG, MARCELLA CORNIA, LORENZO BARALDI, RITA CUCCHIARA, FLORIAN SCHIFFERS

2026

- Developed CounterVid, an agentic VLM and diffusion pipeline that generates targeted counterfactual pairs for object, action, and temporal hallucinations; work conducted at Amazon Prime Video.

Hyperbolic Safety-Aware Vision-Language Models – CVPR 2025 (Highlight)

Nashville, USA

TOBIA POPPI, TEJASWI KASARLA, PASCAL METTES, LORENZO BARALDI, RITA CUCCHIARA

2025

- Introduced HySAC, using hierarchical hyperbolic representations to redirect unsafe retrievals while preserving utility; selected as a **CVPR Highlight** (top ~2.8% of submissions).

Safe-CLIP: Removing NSFW Concepts from Vision-and-Language Models – ECCV 2024

Milan, Italy

SAMUELE POPPI, TOBIA POPPI, FEDERICO COCCHI, MARCELLA CORNIA, LORENZO BARALDI, RITA CUCCHIARA

2024

- Developed a fine-tuning method that dissociates unsafe concepts from vision-language representations for safer retrieval and generation.

Improving LLM First-Token Predictions in MCQA via Output Prefilling – ICPR 2026 (Oral)

Lyon, France

SILVIA CAPPELLETTI, TOBIA POPPI, SAMUELE POPPI, ZHENG-XIN YONG, DIEGO GARCIA-OLANO, MARCELLA CORNIA, LORENZO BARALDI, RITA CUCCHIARA

2026

- Introduced output prefilling, a structured natural-language prefix that steers LLMs toward valid multiple-choice responses, improving accuracy and calibration.

Do Models Share Safety Representations? Cross-Model Steering for Safe Visual Generation – Under Review

TOBIA POPPI, SILVIA CAPPELLETTI, SARA SARTO, FLORIAN SCHIFFERS, GARIN KESSLER, MARCELLA CORNIA, LORENZO BARALDI, RITA CUCCHIARA

2026

- Identified transferable safety directions through cross-model representation analysis across heterogeneous text-to-image and text-to-video generators.

Technical Skills

Research	Responsible Generative AI, AI Safety & Alignment, Multimodal Learning, Vision-Language Models, Diffusion Models, LLMs
ML	PyTorch (advanced), Hugging Face, TensorFlow, JAX (familiar), NumPy, Pandas, Scikit-learn, OpenCV
Engineering	Python, C++, C, Java, Bash, SQL, Docker, Git, Linux/Unix, Slurm, AWS EC2

Fellowship & Research Projects

Amazon Post-Internship Research Fellowship

Seattle, USA / Modena, Italy

PRINCIPAL INVESTIGATOR (STUDENT)

2025 – 2026

- Secured a competitive **\$30,000 research award** (\$20K gift + \$10K compute) and established a funded collaboration between Amazon US Science and Unimore.

European & National AI Projects

Europe / Italy

RESEARCH CONTRIBUTOR

2023 – Present

- Contribute to **ELSA**, **ELIAS**, and **FAIR** projects on safe, robust, sustainable, and multimodal AI; active in the ELLIS network through the ELLIOT project.

Education

University of Pisa – National PhD in AI for Society

Modena, Italy

PHD IN ARTIFICIAL INTELLIGENCE

Nov. 2023 – Nov. 2026 (expected)

- Focus on Responsible Generative AI, safety alignment, robustness, and transparency for multimodal systems; hosted at AlmageLab, University of Modena and Reggio Emilia.

University of Modena and Reggio Emilia

Modena, Italy

MASTER'S DEGREE IN COMPUTER ENGINEERING

Sep. 2021 – Oct. 2023

- 110/110 cum laude**; Erasmus+ exchange at NTNU, Trondheim (Aug. 2022 – Dec. 2022).

University of Modena and Reggio Emilia

Mantova, Italy

BACHELOR'S DEGREE IN COMPUTER ENGINEERING

Sep. 2018 – Oct. 2021

- Grade: 105/110.

Communication & Research Service

Teaching: AI from Zero to Hero, Médecins Sans Frontières (2026); Scalable AI and AI Safety for Generative AI, Unimore MSc (2026); Deep Learning & Computer Vision, Experis (2025).

Reviewing: NeurIPS 2026, ARR, CVPR, ECCV, CVPR Workshops, IEEE TPAMI, Pattern Recognition, and CVIU. **Languages:** Italian (native), English (full professional proficiency).